

VEX Base Bot Quiz - Teacher answer key

After Microsoft Forms Quick Import, open each question, confirm the correct option, set 1 point, and paste the feedback line below.

1. Answer A - V5 Brain

The V5 Brain runs the program and drives ports.

2. Answer B - VEXcode with Blocks or C++ options

VEXcode supports Blocks and text (C++); both upload to the Brain.

3. Answer A - How far a wheel or motor has turned

Encoders count turns so distance can be measured in autonomous.

4. Answer B - Left motor cable, port, and wheel bind

Identical commands still curve if a cable is loose or a wheel binds.

5. Answer B - The stop condition is reading the floor

A floor-facing distance sensor triggers the stop condition early.

6. Answer B - A lower, centred mass improves stability

Centred mass reduces tipping and curved turns.

7. Answer B - V5 Brain

Only the Brain executes the uploaded program.

8. Answer A - Turning by running left and right sides at different speeds

Left/right speed difference turns the drivetrain.

9. Answer B - The robot follows pre-programmed instructions without live driver input

Autonomous uses pre-written code without live driver input.

10. Answer B - Reduces top speed on long runs

More torque usually means less top speed.

11. Answer A - Confirm autonomous and driver programs are downloaded and selected on the Brain

Confirm program slots before rewriting code.

12. Answer B - Weak or inconsistent motor drive

Low voltage often shows up as weak or inconsistent motors.

13. Sample: loose cables make good code fail; check wiring before rewriting loops.

14. Sample: rotation/encoder = turns; distance = range; bumper = contact; IMU = heading.